BCS-5A AHCI - Assignment #2. 24K-0756.

Question 1:

(a) frequency of use.

(b) . This is done to reduce the number of steps required & allow users to access them easily & quickly.
. Fewer opt(s) means faster decisions (Hicks law) & lower cognitive load

(c) Usability improvement:
- Learnability:- A few clear opts are easy for new users to understand.
- Flexibility:- Freq. tasks get a fast path & rare tasks remain reachable through sec. menus.
- Robustness:- The main screen matches what users actually come to do, less clutter means fewer wrong selections

Question 2:

(a)

1. Hidden Coupon Button (Cart ~~Page~~)
Problem: coupon is not visible.
principle violated: Visibility & discoverability.
~~Improvements:~~

2. Order Button (checkout ~~Page~~).
Problem: Primary & secondary actions needs clear visual distinction
Principle violated:- Affordance, error prevention

3. Order Confirmation (Tracking, ~~Page~~).
problem: The initial confirmation screen is not provided. no order confirmed message, summary, order no. etc.
violates: feedback & closure.

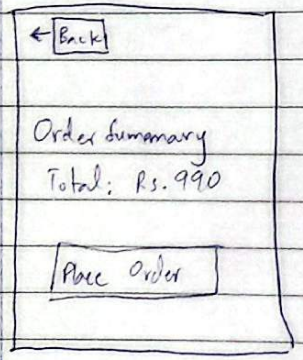
4. Delivery Address Form (checkout) ~~Page~~
problem: Long delivery instructions may be difficult to read in available field
Principle violated: Alignment & error prevention

(b) Article: Julia-Nehme and Rosell, J. (2025) Interaction & Design Barriers for Older Adults in Food Delivery Apps: A Usability study.
Key Finding:
Study identified usability barriers including unclear navigation, insufficient feedback, info. overload, & difficulty identifying interactive elements.
Relations:
This supports making the coupon button easier to find, distinguishing important buttons, providing ~~after~~ order info. confirmation, & simplifying address entry.

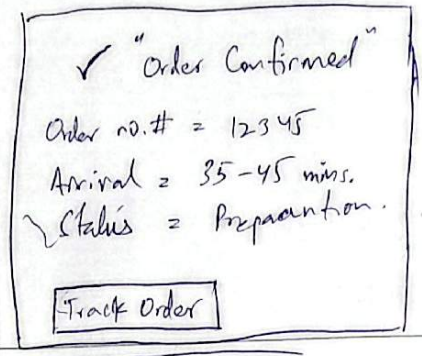
(Q32c) 1. Hidden Coupon Button:-
Improvements:
- Add a clearly labelled 'coupon' field near the 'order total'.
- Use prominent "Apply" button.



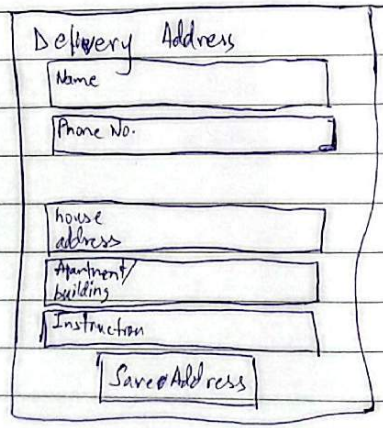
2. Order Button Improvements :-
- Make the 'place order' button larger & prominent.
- Keep it visually seperate from back ~~btn~~ buttons



3. Order Confirmation Improvements:
- Display an 'Order Confirmed' message after successful ordere.
- Show the order no. & estimated delivery time



4. Delivery Address Form Improvement
- Provide larger, multiline text field.
- Use clear labels & constistent spacing.



Question 3:

(a) User research helps designers understand user's abilities, & problems. Without it, EduConnect may fail to meet the different requirements of students, teachers, admin staff, resulting in a difficult-to-use system.

(b) Create 2 2personas:

Person 1: Student.

Attribut	Details	Design Solution :-
Name	Ali	Provide a simple, mobile-friendly dashboard with shortcuts & notifications.
Age	20	
Device	Smartphone	
Exp	Comfortable with tech.	
Goals	Reg. courses, view grades, pay fees, check schedule	
Frustrations	Confusing navigation & too many steps.	

Persona 2: Teacher:-

Attributes:-	Details:-	Design Solution:-
Name	Professor Imran	Provide simple navigation, readable text, & easy marking uploading options.
Age	58	
Device	Desktop/laptop.	
Experience	Basic tech to log in & files	
Goals	Upload marks, share material, mark attendance.	
Frustrations	Slow uploads & complicated menus.	

Step | N' | D(v) | D(w) | p(y) | u(z)

(c) Diary Study: Ask students & teachers to record their daily portal activities & problems. This helps identify

User Photo Diary: As users to photograph their working environments & devices. This helps designers understand the conditions in which they use EduConnect.

(2d) Iterative Design involves repeatedly testing & improving aspects.

Eg 1:
"Test a student dashboard to identify confusing log details"

Eg 2:-
Test the teacher module to identify difficulties in uploading marks and introduce bulk upload

(2e) The Wizard Of Oz technique involves testing a system that appears automated but is partially controlled by a human behind the scenes.

Application:- The university can create a simulated EduConnect app where a researcher manually responds to student access actions.

Benefit:- It allows the team to test the interface & identify usability problems before fully developing the application

(2f) 1. Consistency: Use the same colors, fonts & button styles throughout EduConnect. For eg: keep Save button consistent across all modules.

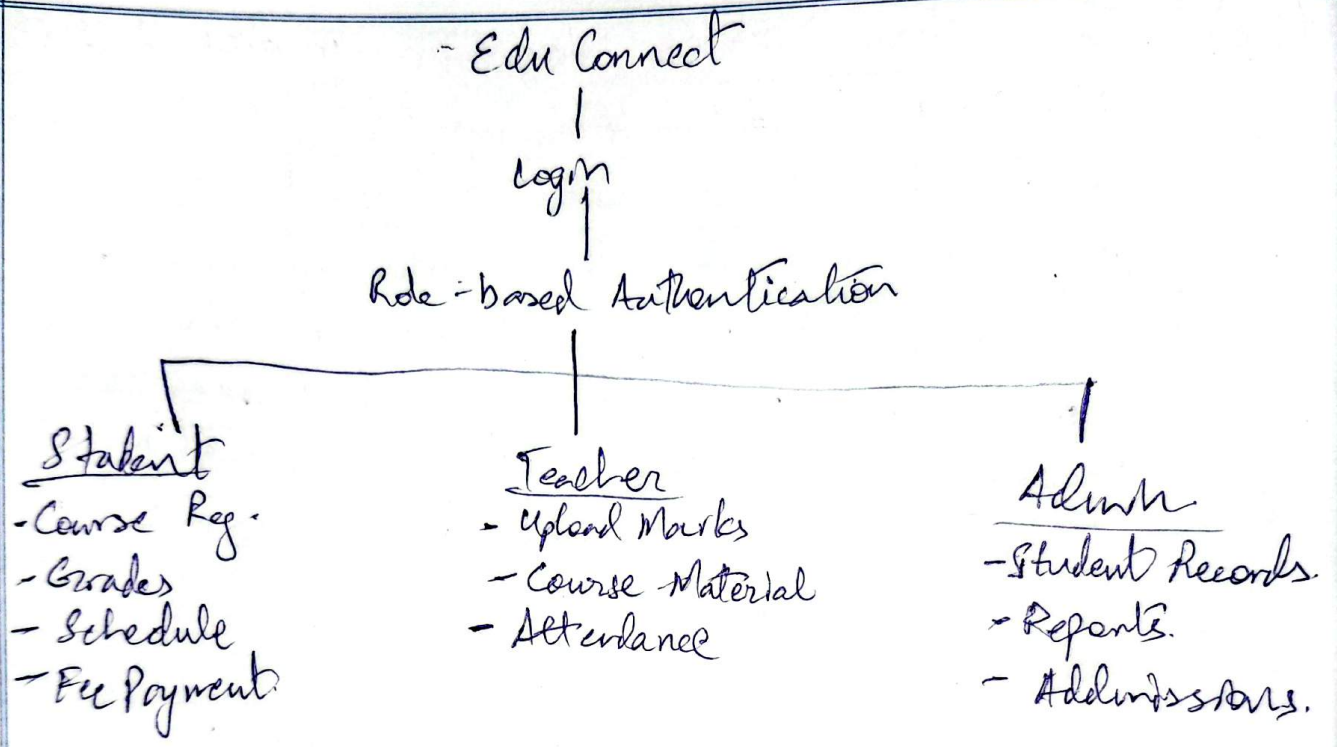
2.

2. Informative feedback:- Display confirmation messages after important actions. for eg:- show "marks uploaded successfully" after a teacher submits grades.

3. Prevent Errors: Validate information & ask for confirmation before important actions. E.g -> show a review screen before final grade submission.

4. Reduce Short-term Memory Load: keep relevant info. visible. E.g -> display selected & available courses together during registration.

(2g)


Edu Connect
Login
Role-based Authentication
Student
- Course Reg.
- Grades
- Schedule
- Fee Payment
Teacher
- Upload Marks
- Course Material
- Attendance
Admin
- Student Records.
- Reports.
- Addmissions.

11:45

foodpanda

A1-102, kaneez fatima block 3,
gulzar-e-hijri, scheme 33, karachi

Up to
50% OFF
on your favourite meals!



Food Groceries Pick-Up Shops More

Popular restaurants

See all

KFC
30–40 min • Free delivery

McDonald's
25–35 min • Free delivery

Home Search Orders Account

CamScanner

5:19 PM

Search for restaurants and groceries

Sort Offers Ratings 4.0+



PRO 25% off selected items +1

Red Apple - Safoora 4.7 (1k+)

From 25 min · $$$ · Pakistani

From Rs.209

Up to 20% off



PRO Free delivery with Rs.699 spend

Burger O'Clock- Gulistan e Johar 4.7 (5k+)

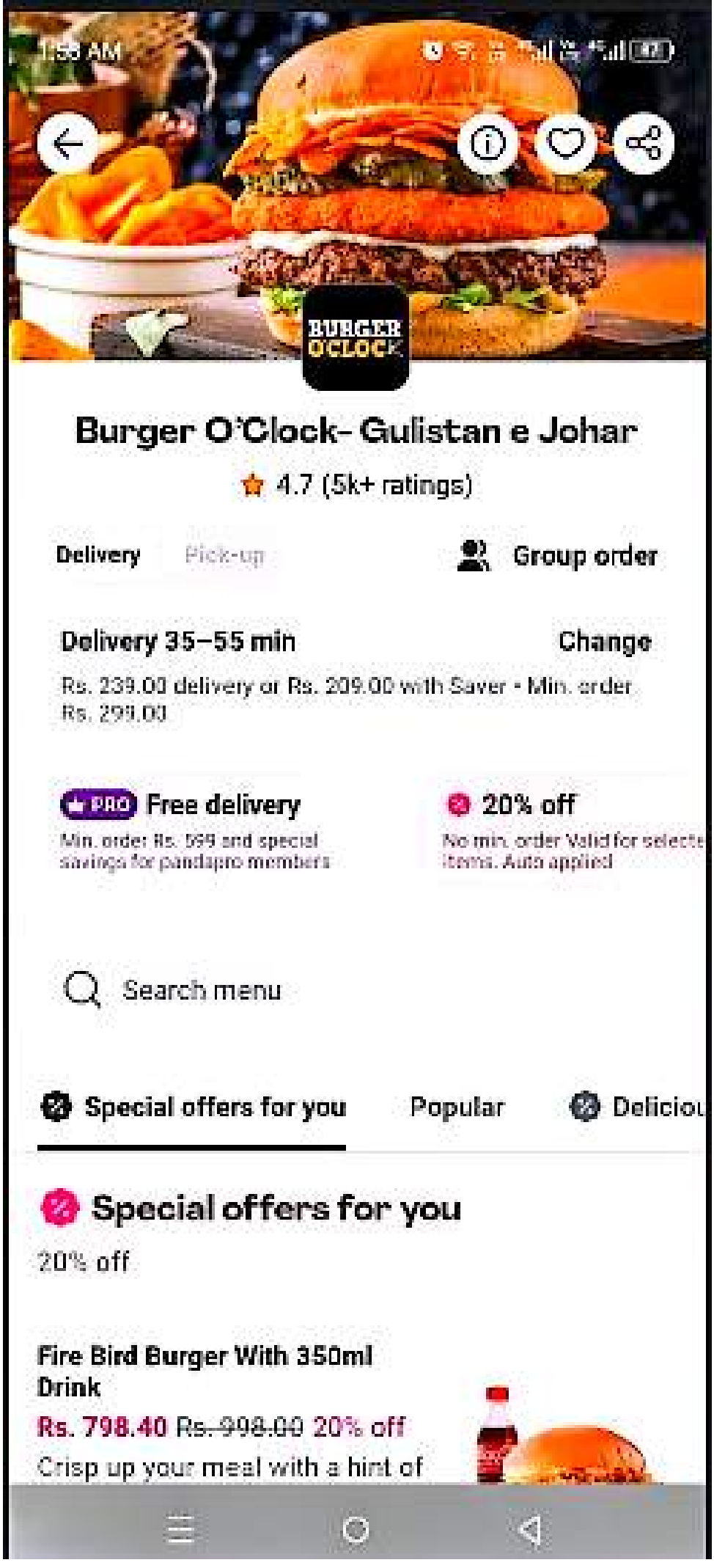
From 35 min · $$$ · Burgers

From Rs.209

20% off selected items



Food Grocery Search Carts Account


1:53 AM
BURGER O'CLOCK
Burger O'Clock- Gulistan e Johar
4.7 (5k+ ratings)
Delivery
Pick-up
Group order
Delivery 35–55 min
Change
Rs. 239.00 delivery or Rs. 209.00 with Saver • Min. order Rs. 299.00
PRO Free delivery
Min. order Rs. 599 and special savings for pandapro members
20% off
No min. order Valid for selecte items. Auto applied
Search menu
Special offers for you
Popular
Deliciou
Special offers for you
20% off
Fire Bird Burger With 350ml Drink
Rs. 798.40 Rs. 998.00 20% off
Crisp up your meal with a hint of

1:53 AM

Cart

Burger O'Clock- Gulistan e Johar

1 Menu — 2 Cart — 3 Checkout

Delivery | Pick-up

Delivery: 40 - 50 min



Beef Smoky Tang With 350ml Drink

Coca Cola - 350 Ml

1

Rs. 1,198.00

+ Add more items

Popular with your order

Other customers also bought these







Total (incl. fees and tax) **Rs. 1,637.69**

See summary

Confirm payment and address

CamScanner



The rider will deliver your order by motorbike

Chat with your rider



CamScanner